MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Discipline Specific Core Course (DSC) - MG1DSCAQC100
INTRODUCTIONTO AQUACULTURE SYSTEMS AND RESOURCES
(2024 ADMISSION ONWARDS)
Duration: 1.5 Hours
Maximum Marks: 50
Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any **10** out of **12** Questions
Each Question carries **2** marks

1. What are the major characteristics of static aquaculture system?

   No Water Flow or Continuous Exchange, Low or Moderate Stocking Density, Because water is not flowing, it cannot dilute wastes effectively, Dependence on Natural Ecological Processes, natural food organisms (plankton), and self-purification processes of the pond ecosystem, Higher Risk of Water Quality Fluctuations, Fertilizers (organic or inorganic) are added to increase natural productivity, Accumulation of Metabolites and Waste, Periodic removal of sludge or drying of ponds between cycles is required, Simple Infrastructure and Low Input System, Often used in traditional aquaculture, rural farms, or low-cost farming systems, Suitable for Hardy Species Eg: Carp, Tilapia, Catfish

2. Discuss the effect of Turbidity on pond productivity

   Turbidity reduces pond productivity by limiting the penetration of sunlight into the water. This results in reduced oxygen levels and diminished natural food availability for fish. High turbidity from suspended clay particles or organic matter can also clog fish gills, stress the fish, and interfere with feeding.

3. Give example for a freshwater prawn. Mention its family

   Macrobrachium rosenbergii, Macrobrachium malcolmsonii, Macrobrachium idella

   Family : Palaemonidae

4. Write any 2 advantages of cage culture

Efficient use of existing water bodies, Easy management and harvesting, low initial investment, high stocking density, better water exchange, easy expansion, flexibility in species cultured (any two)

5. Name the major mechanical filters used in recirculatory aquaculture system

Screen filters / Mesh filters, Drum filters (Rotary Drum Filter), Sand filters (Rapid sand filter / Pressure sand filter), Bead filters

6. Define sewage fed fishculture

Sewage-fed fish culture is an aquaculture practice in which treated or partially treated sewage water is used to grow fish in specially managed ponds. The nutrients present in sewage, mainly organic matter, nitrogen, and phosphorus promote the growth of phytoplankton, zooplankton, and other natural fish food organisms, which support fish production.

7. Suggest any 2 methods of mussel culture

Rack culture, Raft culture, Long line culture

8. Define Weed Fishes. Give example

Weed fishes are small, unwanted, and often fast-breeding fish species that enter culture ponds unintentionally and compete with cultured fish for food and space. They may also prey on fish eggs and fry, thereby reducing the yield of the main cultured species. Example:*Ambassis, Amblypharyngodon, Puntius*

9. Mention any two differences between clams and mussels

Shell Shape, Habitat, Attachment, Shell Surface, Lifestyle Mention any two differences

10. Write down any two benefits of IMTA in aquaculture

Efficient Resource Utilization, Improved Water Quality, Environmental Sustainability, Higher Total Productivity, Economic Stability, Better Ecosystem Balance, Reduced Dependence on External Inputs. Social Acceptance

11. Write down any advantages of Poultry-cum-Fish Culture

Low-cost fertilization, Increased fish yield, Efficient waste utilization,Additional income, Reduced feed cost, Better resource use

12. Expand the acronym GIFT. Write a note on its importance in farming

GIFT – Genetically Improved Farmed Tilapia

The GIFT strain of tilapia is a selectively bred variety developed to improve growth, survival, and overall productivity in aquaculture. It is widely used in many countries because of its superior performance compared to traditional tilapia strains.

Faster Growth Rate, Higher Production, Better Feed Conversion Ratio (FCR), Hardy and Disease-Resistant, Economically Beneficial, Adaptable to Various Culture Systems

Part B

Short Answer Questions

Answer any **3** Questions

Each Question carries **6** marks

13. Explain the advantages and disadvantages of extensive aquaculture

Explain the following Advantages

Low Cost of Production, Environmentally Friendly, Low Management Requirement, Stress-Free Environment for Fish, Low Disease Occurrence

Explain the following Disadvantages

Low Productivity, Limited Control, Slow Growth Rate, Higher Predation Risk, Seasonal Variations Affect Production, Not Suitable for High-Value Species

14.Elaborate upon the diverse chemical soil quality parameters in fish culture Ponds

Explain : Soil pH,Soil Organic Matter , Nitrogen Content, Available Phosphorus, Calcium and Magnesium Levels, Soil Salinity, Soil Acidity , Sulphur Content, Carbonate and Bicarbonate Levels, Redox Potential

15.Distinguish between Tilapia and Etroplus

Explain the mentioned differences for both : Scientific Classification, Species Commonly Found in India, Habitat, Feeding Habit, Reproductive Behaviour, Aquaculture Importance, Body Features

16. List out any six advantages and challenges faced by mariculture practices

Explain Advantages of Mariculture

High Production Potential, Reduces Pressure on Wild Stocks, Faster Growth of Marine Species, Economic Benefits, Efficient Resource Utilization, Diverse Species Culture, Environmental Benefits

Explain Challenges : Environmental Risks, Climatic and Natural Hazards, Disease Outbreaks, High Initial Investment, Conflict Over Coastal Space, Technical Expertise Needed, Regulatory / Legal Challenges, Risk of Escapees

17.Explain the concept of Integrated Farming and discuss its environmental and economic benefits. Provide examples to support your answer

Concept : Integrated Farming is a sustainable agricultural system in which different farm components such as crops, livestock, fish, poultry, biogas, and

horticulture are scientifically linked together so that the waste from one component becomes a resource/input for another.This approach ensures optimal utilization of resources, reduces environmental pollution, and enhances farm productivity.

Benefits : Efficient Resource Recycling, Reduced Use of Chemical Fertilizers, Improved Soil Fertility, Lower Environmental Pollution, Enhanced Biodiversity, Multiple Income Sources, Higher Overall Productivity, Reduced Input Costs, Employment Generation, Better Resilience and Stability

Examples: Rice–Fish–Duck Farming, Poultry–Fish Culture, Crop–Livestock–Biogas System. Explain how any of these integrated farming system benefits

Part **C**
Essay Type Questions
Answer any **1** Question
Each Question carries **12** marks

18. List and define the challenges that aquaculture faces in comparison to agriculture

1. Water Quality Management : Aquaculture requires continuous monitoring and control of parameters such as dissolved oxygen, pH, temperature, ammonia, nitrite, turbidity, and salinity.Even slight fluctuations can stress or kill fish, unlike crops which can tolerate wider environmental variations.
2. Disease Outbreaks : Rapid spread of bacterial, viral, fungal, or parasitic infections in dense stocking conditions.Fish live in a shared water medium; pathogens spread faster than in terrestrial crops or livestock.
3. Dependence on High-Quality Water Resources: Aquaculture systems need reliable sources of clean and sufficient water for filling, exchange, and maintaining pond ecosystems.Agriculture can often depend on rainfall or stored water, while aquaculture fails quickly without adequate water supply.
4. High Sensitivity to Environmental Factors : Aquatic organisms are sensitive to changes in temperature, pollutants, dissolved gases, and salinity.Water transmits and amplifies changes faster than soil or air, making aquaculture systems more vulnerable.
5. Feed Dependency and Cost: Most cultured fish require nutritionally balanced formulated feed.Feed is expensive and constitutes 50–70% of production cost, while many agricultural crops grow using natural soil nutrients and fertilizers.
6. Risk of Escape and Biodiversity Issues: Farmed fish may escape into natural water bodies and compete with or hybridize with wild fish.Agriculture rarely causes such ecological impacts; escapees can disrupt native aquatic ecosystems.
7. Limited Species Domestication: Only a few fish and shellfish species are fully domesticated compared to many crop plants and livestock species in agriculture.Lack of selective breeding limits productivity, growth rate, and disease resistance.
8. High Initial Capital Investment : Construction of ponds, tanks, cages, aeration systems, and water exchange mechanisms requires large initial investment.Many agricultural activities can begin on open land with relatively lower infrastructure costs.
9. Technical and Skill Requirements : Aquaculture demands specialized knowledge of pond management, water chemistry, fish physiology, hatchery technology, and disease control. Agriculture is well-established and familiar to most rural communities, whereas aquaculture is still skill-intensive.

10. Market Fluctuation and Perishability: Fish are highly perishable products requiring cold chain management.Agricultural produce like grains have longer shelf-life, while fish spoil rapidly if not handled properly.
11. Environmental Pollution Risks : Uneaten feed, fecal matter, and chemicals can accumulate and cause eutrophication or toxic conditions.Waterborne pollution spreads quickly, affecting entire production systems and nearby water bodies.
12. Regulatory and Licensing Restrictions : Aquaculture farms often require multiple permits for water use, effluent discharge, land conversion, and species culture.
Agriculture generally faces fewer bureaucratic hurdles.

19.Discuss the structure of fish ponds

The structure of a fish pond refers to the physical layout, components, and design features that support scientific fish culture. A well-designed pond ensures proper water management, good growth of fish, ease of harvesting, and prevention of escape or entry of unwanted organisms.

1. Pond Dykes (Bunds)

Earthen embankments built around the pond.

Provide structural support and prevent water leakage.

Made with well-compacted clay soil.

Inner slopes are gentle (2:1) to prevent erosion.

Outer slopes may be steeper (1.5:1).

Dykes should be strong enough for movement of workers or vehicles.

2. Inlet Structure

Allows controlled entry of water into the pond.

Fitted with screens or filters to prevent unwanted fish, predators, insects, and debris from entering.

Must deliver water without disturbing the pond bottom.

Ideally located at a higher elevation than the outlet for gravity flow.

3. Outlet Structure

Facilitates controlled drainage of water.

Often equipped with a monk structure, sluice gate, or pipe system.

Helps remove bottom water, sludge, or during harvesting.

Should include screens/gratings to prevent escape of stocked fish.

4. Pond Bottom

Should be smooth, level, and gently sloping (1–2% slope) towards the outlet for complete drainage.

Clayey soil is ideal to prevent seepage.

Free from pits, cracks, and hard stones.

Designed to accumulate waste towards the outlet for easy removal.

5. Pond Shape and Size

Common shapes: rectangular or square, as they facilitate netting and harvesting.

Size varies based on purpose:

Nursery ponds: 0.02–0.05 ha

Rearing ponds: 0.1–0.2 ha

Grow-out ponds: 0.4–1.0 ha

Depth is typically 1–1.5 m for common carp and 1.5–2.5 m for species like catla.

6. Water Supply and Drainage System

Must ensure constant water availability.

Should allow water exchange without disturbing fish.

Efficient drainage prevents stagnation and reduces disease risk.

7. Fencing and Protection

Fencing prevents entry of predators like birds, snakes, and mammals.

Bamboo screens, nylon nets, or wire fencing may be used.

Watchtowers or guard huts for large farms.

8. Peripheral and Internal Canals

Some advanced ponds have canals for:

Water circulation

Collection of sludge

Movement of fish during harvest

9. Sluice and Monk Structures

Sluice gate: A small channel with gates to control water flow.

Monk: Vertical concrete structure with slots for boards to regulate pond water depth and drainage.

Allows selective removal of surface or bottom water.

10. Additional Functional Structures

a. Spillway

Prevents overflow during heavy rains by releasing excess water.

b. Aeration System

Mechanical aerators or paddle wheels to maintain dissolved oxygen.

c. Sedimentation or Supply Canal

Removes silt before water enters the main pond.

d. Harvesting Pit (Collection Pit)

Depressed area near the outlet where fish accumulate during drainage, enabling easy harvest.

20. Write an essay on the taxonomy and morphological features of cold water fishes and sport fishes

Cold-water fishes constitute an important group of finfishes that inhabit the cool, fast-flowing streams, rivers, and lakes of temperate and sub-temperate regions. Many of these fishes are highly valued for their food quality, growth characteristics, and recreational importance, thus forming an essential component of sport fisheries.

Taxonomy of Cold-Water Fishes

Cold-water fishes belong primarily to the orders Salmoniformes, Cypriniformes, and Perciformes

1. Order: Salmoniformes

This order includes the true cold-water species known for their migratory behaviour and high economic value.

Family: Salmonidae

Genera: *Salmo, Oncorhynchus, Salvelinus*

Important species:Brown trout (*Salmo trutta*), Rainbow trout (*Oncorhynchus mykiss*), Brook trout (*Salvelinus fontinalis*)

2. Order: Cypriniformes

Many snow-fed Himalayan streams harbour cyprinid cold-water fishes.

Family: Cyprinidae

Genera: *Tor, Schizothorax*

Important species:

Mahseer (*Tor putitora, Tor khudree*), Snow trouts (*Schizothorax spp.*)

Mahseer is one of the most prestigious game fishes of India, known for its strength and size.

3. Order: Perciformes

Certain cold-water perches and related fishes are also popular with anglers.

Family: Percidae, Serranidae, or Cichlidae (depending on region)

These fishes inhabit cool lakes and reservoirs and exhibit strong predatory traits.

Morphological Features of Cold-Water Fishes

Cold-water fishes possess several morphological adaptations that enable them to survive in low temperatures and fast-flowing waters.

1. Body Shape

Generally streamlined, fusiform bodies help reduce friction during swimming in swift currents.

Trout and mahseer show powerful, muscular bodies for sustained swimming.

2. Colouration

Cold-water fishes often display silvery, dark-mottled, or spotted patterns, providing camouflage in clear waters.

Rainbow trout exhibits a characteristic pink lateral band.

3. Fins

Strong, well-developed paired pectoral and pelvic fins for maneuverability in flowing water.

Caudal fin is usually forked, aiding speed.

In Salmonids, the presence of a small adipose fin between the dorsal and caudal fin is a key diagnostic feature.

4. Gill Structure

Large gill surface area promotes efficient oxygen uptake in cold water, which naturally holds high dissolved oxygen.

5. Scales

Most species have cycloid scales that are smooth and flexible, aiding rapid movement.

Snow trouts have smaller scales adapted to rocky habitats.

6. Mouth and Feeding Adaptations

Salmonids possess a terminal mouth with sharp teeth for carnivorous feeding.

Mahseer shows a sub-terminal mouth with thick, fleshy lips for scraping food from rocks.

7. Swim Bladder

Well-developed for buoyancy control

Sport Fishes: Taxonomy and Characteristics

Sport fishes are species that are targeted by anglers for recreational fishing. While many are cold-water fishes, not all sport fishes necessarily inhabit cold environments. Sport fishes are identified based on fighting ability, size, speed, and overall challenge to anglers.

1. Trout and Salmon (Order: Salmoniformes)

Renowned worldwide for sport fishing.

Their strong, energetic movement and ability to jump make them ideal for angling.

2. Mahseer (Order: Cypriniformes)

Considered the Tiger of Indian rivers

Exhibits explosive strength, making it a prized game fish.

Morphological Features of Sport Fishes

Sport fishes generally possess traits that contribute to their agility, strength, and endurance.

1. Strong Muscular Body

    Well-developed muscles support powerful bursts of speed needed to escape predators and resist capture.

2. Well-Developed Caudal Peduncle

    A narrow but sturdy caudal peduncle increases thrust during fast swimming.

3. Fin Arrangement

    Large dorsal and anal fins for stability.

    Forked caudal fins for speed and endurance.

4. Streamlined Body for Speed

    Minimizes drag and allows rapid directional changes when hooked.

5. Predatory Mouth Parts

    Many sport fishes have sharp teeth enabling them to catch prey actively.

6. Strong Dentition

    Helps grasp live prey, making them aggressive and challenging game species.

7. Brilliant Colouration

    Many sport fishes exhibit attractive coloration, adding to their ornamental and recreational appeal.